

SDR-Based Spectrum Monitoring System for FM Broadcast Compliance Assessment

Continuous Monitoring of FM Broadcast Using HackRF One

Project Design Document

September 21, 2026

1 Introduction

Objective

Design, build, and field-validate a continuously operating spectrum-sensing sensor that monitors FM broadcast services and records per-channel received power and occupancy over time. The system must be low-cost, reproducible, and scalable from one sensor to a network of geographically distributed sensors reporting to a central database.

System Overview

The sensor chain is:

*Antenna → HackRF One → GNU Radio DSP → measurement writer → time-series DB
→ web dashboard*

Figure 1 shows the end-to-end architecture for both the single-sensor (Phases 0–5) and multi-sensor (Phase 6) configurations.

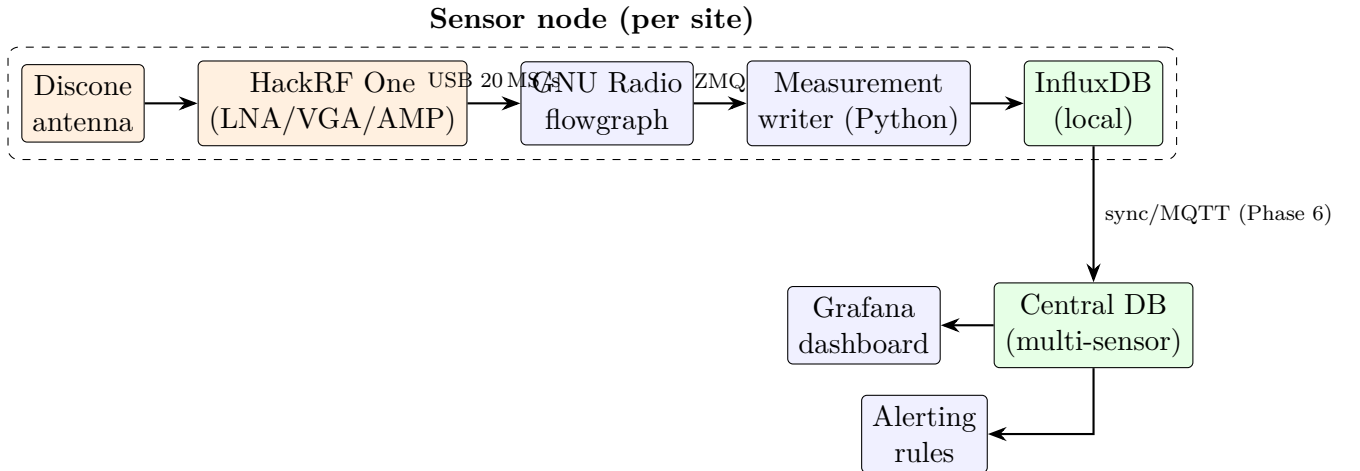


Figure 1: End-to-end sensor architecture. Dashed box: one sensor site; solid nodes below the dashed box: central services activated in Phase 6.

Scope

- **In scope:** sensing and logging of FM broadcast occupancy and power; calibration against references; web visualization; multi-sensor aggregation.
- **Out of scope (v1):** demodulation/recording of audio content, transmitter geolocation, real-time regulatory enforcement actions.

2 Phase 0 — Requirements, Measurand Definition, and Calibration

Measurand Definition

The primary measurands are defined per FM channel c with centre frequency f_c and channel bandwidth B_c :

Table 1: Measurand definition.

Quantity	Symbol	Unit	Definition
Channel power	P_c	dBm	Integrated power over $B_c = 200$ kHz
Noise floor	P_{nf}	dBm/Hz or dBmin B_c	Median power of vacant channels
Occupancy flag	o_c	boolean	$P_c > \gamma$ (adaptive threshold)
Duty cycle	D_c	%	Fraction of time $o_c = 1$ over a window
Estimated field strength	E_c	dB μ V/m	From P_c via antenna factor
Full-band scan timestamp	t	UTC	Start of scan, ms resolution

System requirements are summarized in Table 2.

Table 2: Top-level requirements.

ID	Requirement	Target
R1	Frequency range	87.5–108 MHz
R2	Full-band scan period	≤ 10 s
R3	Occupancy detection	$P_{fa} \leq 5\%$, $P_d \geq 95\%$ at $\text{SNR} \geq 0$ dB
R4	Absolute power accuracy	± 3 dB (post-calibration)
R5	Frequency accuracy	± 5 kHz at 100 MHz
R6	Uptime	$\geq 95\%$ over a 30-day run
R7	Scalability	≥ 5 sensors feeding one central DB

HackRF One Hardware Characterization

Gain chain: The HackRF receiver path provides three digitally controlled gain stages: RF amplifier (AMP, 0/14 dB, fixed step), LNA (0–40 dB, 8 dB steps), and VGA (0–62 dB, 2 dB steps). Sensing uses **fixed** gains — automatic gain control must remain disabled because AGC destroys absolute-power traceability.

Recommended starting point for a moderate-signal urban/rooftop site: RF amp off (0 dB), LNA = 16 dB, VGA = 20 dB, then adjust per site using the overload checks in Section 9.

Noise figure and dynamic range: The noise figure is estimated in bench tests using the *gain method*:

$$NF \approx P_{n,\text{out}} (\text{integrated in } B) - (-174 + 10 \log_{10} B) - G_{\text{total}} \text{ [dB]}, \quad (1)$$

where $P_{n,\text{out}}$ is the measured output noise power with the input terminated in 50Ω , B the measurement bandwidth, and G_{total} the total applied gain. Target: usable dynamic range ≥ 60 dB with compression no lower than -30 dBm at the antenna port.

Frequency accuracy. The stock crystal is specified at roughly ± 20 ppm; at 100 MHz this is up to ± 2 kHz. This is acceptable for FM channel identification (R5) but the offset must be measured against a known station or reference and stored per device. A TCXO modification or GPSDO is optional and only needed for cross-sensor interference studies.

Power Calibration: HackRF reports uncalibrated digital power; a per-frequency calibration offset converts raw measurements to dBm:

$$P_c [\text{dBm}] = P_{c,\text{raw}} + C_{\text{cal}}(f_c), \quad (2)$$

where $C_{\text{cal}}(f)$ is obtained at $N \geq 5$ frequencies spanning the band, either from a calibrated signal generator or by comparing against a reference receiver (spectrum analyzer, or an RTL-SDR cross-calibrated against a known field). Interpolate linearly between points (in dB). Record the calibration table per device and firmware revision.

Field-Strength Estimation: With a calibrated antenna factor $AF(f)$ for the monitoring antenna:

$$E [\text{dB}\mu\text{V/m}] = P_r [\text{dBm}] + 107 + AF(f) [\text{dB/m}], \quad (3)$$

since a 0 dBm signal in a 50Ω system corresponds to $107 \text{ dB}\mu\text{V}$. $AF(f)$ is taken from the antenna datasheet or measured in an open-area test against a reference field.

3 Phase 1 – Sensor Build (HackRF One + GNU Radio)

Flowgraph. A capture-based flowgraph samples a 20 MHz band per capture; channel selection, filtering, and power integration happen in the DSP stage (Phase 2). A full-band scan uses two captures (Table 3) to cover the FM band. Figure 2 shows the block topology.

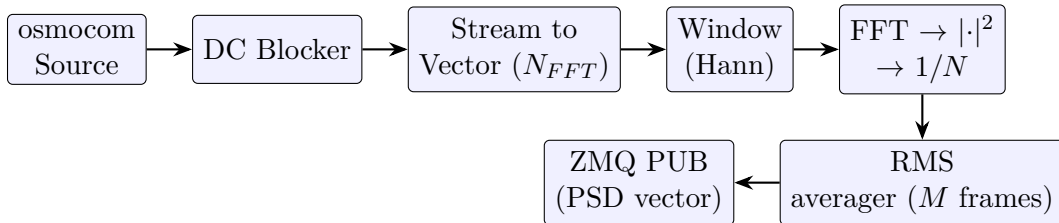


Figure 2: Phase 1 GNU Radio flowgraph topology.

Key source parameters are fixed at build time (Table 3).

A full-band scan comprises two captures. Their union (83–113 MHz) covers the FM band 87.5–108 MHz with margin. The flowgraph retunes between captures or uses two parallel source branches.

Table 3: HackRF source parameters (Phase 1).

Parameter	Value	Note
Sample rate	20 MS/s	Per capture
Capture 1 centre	93 MHz	Covers 83–103 MHz
Capture 2 centre	103 MHz	Covers 93–113 MHz
Bandwidth	20 MHz	HackRF baseband filter
LNA gain	16 dB	Fixed (site-tunable)
VGA gain	20 dB	Fixed (site-tunable)
RF amp	Off (0 dB)	Enable only for weak-signal sites

Bench Bring-up Checklist

1. `hackrf_info` and `hackrf_transfer -r` CLI smoke test (driver and USB integrity).
2. Flowgraph runs 60 s without underruns (UuUu counters zero).
3. Log underrun/overrun counters per full-band scan in the writer for watchdog use (Section 9).

4 Phase 2 — DSP Pipeline and Validation

Signal Processing Chain

PSD estimation. The averaged periodogram over M FFT frames is

$$\hat{S}_{xx}(k) = \frac{1}{M} \sum_{m=1}^M \frac{1}{N_{FFT} U} \left| \sum_{n=0}^{N_{FFT}-1} w(n) x_m(n) e^{-j2\pi kn/N_{FFT}} \right|^2, \quad U = \frac{1}{N_{FFT}} \sum_n w^2(n), \quad (4)$$

with Hann window $w(n)$ and $N_{FFT} = 8192$ (≈ 2.44 kHz bin width at 20 MS/s). $M = 20$ gives a PSD update time of ≈ 8 ms per capture; with the dual-capture full-band scan the scan period remains well below R2.

Channel power integration. Channel power is the band-integrated PSD, not a peak pick:

$$P_c [\text{dBm}] = 10 \log_{10} \left(\sum_{k \in \mathcal{K}_c} \hat{S}_{xx}(k) \Delta f \right) + C_{\text{cal}}(f_c), \quad \Delta f = \frac{f_s}{N_{FFT}}, \quad (5)$$

where $\mathcal{K}_c$ spans the channel allocation $f_c \pm 100$ kHz.

Occupancy detection An **adaptive energy detector** sets the threshold from the local noise floor using robust statistics (median and median absolute deviation):

$$\gamma = \text{median}_{c \in \mathcal{V}}(P_c) + k \cdot 1.4826 \cdot \text{MAD}_{c \in \mathcal{V}}(P_c), \quad o_c = \mathbb{1}\{P_c > \gamma\}, \quad (6)$$

where $\mathcal{V}$ is the set of channels classified as vacant in the previous scan and $k \approx 4$ –6 is tuned in Phase 2 validation.

Upgrade path: cyclostationary detection FM broadcasts carry a 19 kHz stereo pilot, producing strong spectral correlation peaks at cyclic frequencies $\alpha = \pm 19$ kHz. An SCD (spectral correlation density) test on candidate channels discriminates true FM emissions from noise, interference, or intermodulation products — implemented as a Phase 5/6 refinement to reduce false alarms near strong transmitters.

Station identification (ground truth). Optional RDS demodulation (GNU Radio RDS blocks or the `redsea` CLI on narrowband-recorded IQ) extracts Program Identification (PI) codes and station names, giving positive station identification for validation in Phase 5.

Validation Against Known References

Test signals. • **T1:** Recorded FM IQ replayed through a second SDR or RF playback path (known power, known frequency).

- **T2:** A locally receivable station with published parameters (frequency, ERP, site coordinates).
- **T3:** An independent reference receiver (RTL-SDR or spectrum analyzer) logging the same band.

Detector performance. From N annotated scans against ground truth, build the confusion matrix and compute

$$P_d = \frac{TP}{TP + FN}, \quad P_{fa} = \frac{FP}{FP + TN}. \quad (7)$$

Acceptance: R3 ($P_d \geq 95\%$, $P_{fa} \leq 5\%$ at $\text{SNR} \geq 0$ dB).

Power and frequency accuracy • Frequency: detected carrier vs. published frequency; acceptance ± 5 kHz (R5).

- Power: difference vs. reference receiver across ≥ 10 channels; acceptance ± 3 dB (R4), per Section 2.

Plausibility check via link budget. For station T2, expected received power:

$$P_r = \text{EIRP} - \text{FSPL}(d, f) - L_{\text{clutter}}, \quad \text{FSPL} [\text{dB}] = 20 \log_{10}(d_{\text{km}}) + 20 \log_{10}(f_{\text{MHz}}) + 32.44. \quad (8)$$

Agreement within ± 10 dB (allowing clutter uncertainty) confirms the calibration chain end-to-end.

5 Phase 3 — Measurement Database

Technology Choice

InfluxDB 2.x is the default: native time-series storage, downsampling (retention policies), and direct Grafana integration. PostgreSQL/TimescaleDB is an acceptable alternative if relational sensor metadata must be colocated.

Table 4: InfluxDB measurements (line-protocol tags/fields).

Measurement	Tags	Fields
channel_power	sensor_id, freq_mhz, location	power_dbm, field_dbvm, occupied (bool), noise_floor_dbm
spectrum	sensor_id	psd (float array / blob sidecar) — optional waterfall history
health	sensor_id	underruns, gain_lna, gain_vga, temp_c, uptime_s

Schema

Writer Service

A small Python daemon subscribes to the flowgraph’s ZMQ PUB socket, computes per-channel metrics (Phase 2 math lives here, keeping the flowgraph thin), and batch-writes line protocol (batch size 100, flush 1 s). It also emits the **health** measurement every full-band scan and includes a reconnect/backoff on ZMQ or DB failure.

6 Phase 4 — Web Dashboard

Grafana Panels

1. **Occupancy heatmap** (primary view): frequency $\times$ time, colour = channel power. This is the key broadcast-monitoring display.
2. **Live spectrum trace**: current PSD with channel markers and RDS-derived station labels.
3. **Per-channel trends**: 24 h power and duty cycle for selected stations.
4. **Sensor table**: online status, last scan age, noise-floor drift, underrun counts (multi-sensor ready).

Alerting Rules

- Channel silent: **occupied** false for $T_{\text{on-air}}$ hours on a station previously $> 90\%$ duty cycle.
- Out-of-band/anomalous emission: power above threshold outside licensed raster channels.
- Sensor offline: no **health** point for 3 scan periods.
- Noise-floor drift: median noise floor shifts by > 6 dB over 24 h (possible gain fault or interference).

7 Phase 5 — Field Validation

Protocol

1. Select a site with a published FM station list (regulator data: frequency, ERP, transmitter coordinates).
2. Run the sensor continuously for 72 h (satisfies R6 partially; extend to 30 days for full R6).
3. Independently log the band with the reference receiver (T3) for at least 6 h overlapping the deployment.
4. Collect RDS identifications where possible for ground truth.

Metrics and Acceptance

- **Detection rate:** licensed stations detected / licensed stations expected at the site (an ERP-and-distance filter defines “expected”). Target $\geq 90\%$ of expected stations.
- **Ranking consistency:** Spearman’s ρ between measured power ranking and predicted ranking (EIRP – FSPL). Target $\rho \geq 0.8$.
- **Stability:** no unrecovered crashes; watchdog restart count logged; disk usage within budget (see Section 8).
- **Error sources documented:** multipath fading (slow power variation of ± 10 dB is normal for a fixed link over a day), intermodulation from strong FM signals (check phantom products at $2f_1 - f_2$ combinations), and front-end overload (add attenuation and re-run if observed).

8 Phase 6 — Multi-Sensor Deployment

Architecture

Sensor nodes forward data to a central collector via MQTT (topic per sensor: `spectrum/<sensor_id>/channel`) with the telegraf/agent or the writer publishing directly over TLS. The central instance runs InfluxDB and Grafana; dashboards aggregate all sensors using the `sensor_id` tag.

Time Synchronization

Cross-sensor comparison requires aligned timestamps:

- Minimum: NTP (Chrony), keeping nodes within tens of ms.
- For interference/correlation studies: GPS PPS or a shared GPSDO; note scan start times with sub-ms precision.

Deployment

Each node ships as a `docker-compose` stack: `flowgraph` + `writer` + `telegraf` + (`node exporter`). Per-node configuration file holds sensor ID, location, calibration table reference, and fixed gains — identical images across sites, configuration via environment.

Storage budget: at $206 \text{ channels} \times 4 \text{ fields} \times 1 \text{ full-band scan}/5 \text{ s}$, raw `channel_power` volume is approximately 60 MB/day/sensor for 4-byte fields and 120 MB/day/sensor for 8-byte fields, before InfluxDB overhead. A 7-day raw retention with 1 h downsampling beyond 7 days keeps long-term storage negligible. Recalculate for site-specific channel count and scan rate.

9 Risk Register

Table 5: Top risks and mitigations.

Risk	Severity	Mitigation
Absolute power inaccuracy (HackRF uncalibrated)	High	Per-device calibration table (Sec. 2); fixed gains; periodic re-check against a known station (T2).
Front-end overload near transmitters	High	Monitor for intermod products at $2f_1 - f_2$; reduce LNA/VGA; add external attenuator; AMP stays off by default.
USB dropouts on long runs	Medium	Log underruns in <code>health</code> ; watchdog (<code>systemd Restart=always</code>); alert on missing health points.
Frequency drift beyond R5	Low	Measure per-device ppm offset; log drift trend in <code>health</code> ; optional TCXO upgrade.
Noise-floor mis-estimation causing P_{fa} violations	Medium	MAD-based adaptive threshold; cyclostationary confirmation (Phase 5/6).
Database disk growth	Low	Retention policies and downsampling (Sec. 8).

10 Phase-Gate Milestones

Terminology

- *FFT frame*: one N_{FFT} -sample block.
- *PSD update*: averaged periodogram over M FFT frames.
- *Capture*: one contiguous IQ block at a single centre frequency.
- *Full-band scan*: one complete measurement across the FM band (one or more captures, depending on coverage).
- *Scan period*: time between consecutive full-band scans.

References

- [1] ITU-R Recommendation BS.412, *Planning standards for FM sound broadcasting at VHF*.

Phase	Gate	Exit criteria
0	Spec & calibration signed off	Measurand table approved; $C_{\text{cal}}(f)$ measured
1	Sensor online	60 s clean run, zero underruns, ZMQ streaming
2	DSP validated	R3, R4, R5 verified on T1–T3
3	Logging live	≥ 1 h of queryable DB data
4	Dashboard live	Heatmap + alerts firing in Grafana
5	Field validated	Detection rate and ρ targets met; 72 h continuous run with no unrecovered failures. Full R6 (30-day uptime) verified by post-gate soak.
6	Network live	≥ 2 sensors aggregated centrally; full R7 (≥ 5 sensors) verified during network roll-out.

- [2] ITU-R Recommendation SM.1268, *Method of measuring the performance of spectrum analysers*.
- [3] IEC 62106, *Radio data system (RDS) for VHF/FM sound broadcasting*.
- [4] IEEE 802.22, *Sensing methods overview; energy and cyclostationary detection*.
- [5] Great Scott Gadgets, *HackRF One documentation*.